

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        HashSet<Student> set = new HashSet<>();  
        HashMap<Student, HashSet<String>> map = new HashMap<>();
```

```
        Student s1 = new Student("1", "HARSHA");  
        Student s2 = new Student("2", "kewin");  
        set.add(s1);  
        set.add(s2);
```

```
        HashSet<String> c1 = new HashSet<>();  
        c1.add("Java");  
        c1.add("DBMS");
```

```
        HashSet<String> c2 = new HashSet<>();  
        c2.add("Java");  
        c2.add("DSA");
```

```
        map.put(s1, c1);  
        map.put(s2, c2);
```

```
        Iterator<Student> it = set.iterator();  
        while (it.hasNext()) {
```

```
Student s = it.next();  
System.out.println(s.getId() + " " + s.getName());
```

```
Iterator<String> it2 = map.get(s).iterator();  
while (it2.hasNext()) {  
    System.out.println(it2.next());  
}  
}  
}
```

```
class Student {  
    String id, name;  
    Student(String id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
    String getId() { return id; }  
    String getName() { return name; }
```

```
    public boolean equals(Object o) {  
        if (this == o) return true;  
        if (!(o instanceof Student)) return false;  
        Student s = (Student) o;  
        return id.equals(s.id);  
    }
```

```
    public int hashCode() {  
        return id.hashCode();  
    }  
}
```

The screenshot shows a web browser window with the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" and the user is logged in as "R HARSHA KYLE". The assignment is titled "CO2-AT1-Scenario Based Program" and is worth 10 marks. The question is "Q2 E-Commerce Cart System". The problem statement is: "Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order. (a) List three Collection classes applicable for cart management. (b) Represent how cart items are stored using a List-based structure. (c) Write a Java program using ArrayList and Iterator to: • Add items • Remove an item • Display all items". The "Your Code" section shows the following Java code:

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3
4 public class Main {
5     public static void main(String[] args) {
6         ArrayList<String> cart = new ArrayList<>();
7         cart.add("Butter");
8         cart.add("Bread");
9         cart.add("Eggs");
```

The "Input" section shows "stdin input" and the "Output" section is empty. The bottom status bar shows the temperature as 35°C and the time as 03:38 PM on 26-08-2026.

```
import java.util.ArrayList;
import java.util.Iterator;
```

```
public class Main {
    public static void main(String[] args) {
        ArrayList<String> cart = new ArrayList<>();
```

```
        cart.add("Butter");
        cart.add("Bread");
        cart.add("Eggs");
```

```
        Iterator<String> it = cart.iterator();
        while (it.hasNext()) {
            String item = it.next();
            System.out.println(item);
        }
    }
}
```

}

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the lab name, and a user profile for R HARSHA KYLE. The main content area is titled "CO2-AT1-Scenario Based Program" and features a "Library Management System" question worth 10 marks. The question asks for three Map implementations, a schema-like representation, and a Java program using HashMap. The code editor shows a Java program that uses a HashMap to store book details. The sidebar on the left lists five questions, with the first four marked as completed. The bottom status bar displays system information like temperature, time, and date.

```

import java.util.HashMap;

public class Main {

    public static void main(String[] args) {
        HashMap<String, String> books = new HashMap<>();
        books.put("1", "Java Basics");
        books.put("2", "Data Structures");
        books.put("3", "Algorithms");
        String isbn = "1";
        String book = books.get(isbn);
        System.out.println("Retrieved book for ISBN " + isbn + ": " + book);
        System.out.println("All books:");
        for (String key : books.keySet()) {
            System.out.println("ISBN: " + key + " Book: " + books.get(key));
        }
    }
}

class Library {
    HashMap<String, String> books;

    Library() {
        books = new HashMap<>();
    }

    void addBook(String isbn, String name) {
        books.put(isbn, name);
    }
}

```

```
String getBook(String isbn) {
return books.get(isbn);
}

}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and a user profile for R HARSHA KYLE. The main section is titled "CO2-AT1-Scenario Based Program" and contains a list of questions on the left and a detailed problem statement on the right. The problem statement, "Q4 Employee Data Processing," asks for a Java program using an Iterator to process employee records. Below the problem statement, there is a "Your Code" section with a text editor showing a partial Java code snippet. To the right of the code editor are "Input" and "Output" fields. The bottom of the screen shows a Windows taskbar with various application icons and system status information.

```
import java.util.*;
```

```
class Employee {
private int id;
private String name;
private double salary;
```

```
public Employee(int id, String name, double salary) {
this.id = id;
this.name = name;
this.salary = salary;
}
```

```
public int getId() {
return id;
}
```

```
public String getName() {
return name;
```

```
}
```

```
public double getSalary() {  
    return salary;  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        List<Employee> empList = new ArrayList<Employee>();  
        empList.add(new Employee(1, "Asha", 60000));  
        empList.add(new Employee(2, "Ravi", 45000));  
        empList.add(new Employee(3, "Meena", 75000));  
        empList.add(new Employee(4, "Karthik", 50000));  
  
        Iterator<Employee> it = empList.iterator();  
        while (it.hasNext()) {  
            Employee e = it.next();  
            if (e.getSalary() > 50000) {  
                System.out.println(e.getId() + " " + e.getName() + " " + e.getSalary());  
            }  
        }  
    }  
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the lab name, and a user profile for R HARSHA KYLE. The main content area is titled "CO2-AT1-Scenario Based Program" and features a "Questions" sidebar on the left with five items, each marked as completed. The main question, "Online Voting System," is worth 10 marks and asks for a design and Java implementation for an online voting system. Below the question, there is a "Your Code" section with a text area containing Java code, an "Input" field for stdin input, and an "Output" field. The bottom of the screen shows a Windows taskbar with various application icons and system status information.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q5 Online Voting System 10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         String[] inputs = {  
6             "v1", "Alice",  
7             "v2", "Bob",  
8             "v3", "Alice",  
9             "v1", "Bob",  
10            "v4", "Charlie"  
11        }  
12    }  
13 }
```

Input

stdin input

Output

```

import java.util.*;

public class Main {
    public static void main(String[] args) {
        String[] inputs = {
            "V1", "Alice",
            "V2", "Bob",
            "V3", "Alice",
            "V1", "Bob",
            "V4", "Charlie"
        };

        int n = inputs.length / 2;
        Set<String> voted = new HashSet<>();
        Map<String, Integer> count = new HashMap<>();

        for (int i = 0; i < n; i++) {
            String voterId = inputs[i * 2];
            String candidate = inputs[i * 2 + 1];

            if (voted.contains(voterId)) {
                System.out.println("Already voted: " + voterId);
                continue;
            }

            voted.add(voterId);
            count.put(candidate, count.getOrDefault(candidate, 0) + 1);
        }

        System.out.println("\nVoting Results:");
        for (String c : count.keySet()) {
            System.out.println(c + " - " + count.get(c));
        }
    }
}

```